

Institute of Information technology
Jahangirnagar University
ICT 3101 – Operating System
Class Test : 01

Marks: 10

Time : 30 minutes

1. What is the purpose of interrupts? How does an interrupt differ from a trap? Discuss the interrupt-driven I/O cycle with figure. 03
2. Identify several advantages and several disadvantages of open-source operating systems. Identify the types of people who would find each aspect to be an advantage or a disadvantage. 02
3. What is Direct Memory Access (DMA), and how does it improve system performance by reducing CPU intervention during data transfers between high-speed I/O devices and main memory? 02
4. What is the main advantage of the microkernel approach to system design? How do user programs and system services interact in a microkernel architecture? What are the disadvantages of using the microkernel approach? 03

Medulla

1. Describe different pieces of information associated with the PCB. Draw the figure how CPU switches from process to process. Illustrate two models of IPC. 02 3 5
2. What are two differences between user-level threads and kernel-level threads? Under what circumstances is one type better than the other? 02 2
3. Consider the following set of processes, with the length of the CPU burst time given in milliseconds: 04

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8-2-2	4
P4	4-2-1	2
P5	5-2-2-1	3

- The processes are assumed to have arrived in the order P1, P2, P3, P4, P5, all at time 0.
- a. Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2).
 - b. What is the turnaround time of each process for each of the scheduling algorithms in part a?
 - c. What is the waiting time of each process for each of these scheduling algorithms?
 - d. Which of the algorithms results in the minimum average waiting time (over all processes)?

Marks: 10

Time : 30 minutes

3. A small computer lab has 4 computers and 3 types of resources (RAM, CPU cores, Storage). 05

	Allocation Matrix			Max Matrix			Available Matrix		
	RAM(GB)	CPU Cores	Storage(GB)	Max RAM	Max CPU	Max Storage(GB)	RAM(GB)	CPU Cores	Storage(GB)
P1	3	2	50	6	4	100	8	6	150
P2	2	1	30	4	2	60			
P3	1	3	20	3	5	40			
P4	4	2	40	5	3	80			

- Determine if the system is in a safe state
 - Create a safe sequence of resource allocation
 - Verify if all processes can be completed without deadlock.
4. Explain Lock variable mechanism . Does Lock variable met the four conditions of Synchronization mechanism ? Explain, how this problem can be solved ?

Marks: 10

Time: 30 minutes

1. What is the cause of thrashing? How does the system detect thrashing? Once it detects thrashing, what can the system do to eliminate this problem? 04
2. How many page faults occur for First In First Out (FIFO) and Least Recently Used (LRU) algorithm for the following reference string with four-page frames? What will be the miss ratio?
1 2 3 1 5 3 4 5 4 7 8 7 8 9 7 8 9 5 4 5 4 2 04
3. Define (i) Seek Time (ii) Rotational Latency (iii) Disk Access Time (iv) Disk Response Time. 02

Marks: 10

Time: 25 minutes

1. What is page fault? Discuss the steps in handling a page fault? What Happens if there is no Free Frame? 04
2. How many page faults occur for First In First Out (FIFO) and Least Recently Used (LRU) algorithm for the following reference string with four-page frames? What will be the hit ratio? 04
1 2 3 4 5 3, 4 1 6 7 8 7 8 9 7 8 9 5 4 5 4 2
3. Explain SSTF with its advantages and disadvantages. 02

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